PGMNet: A Prototype-Guided Multimodal Network for Ship Recognition in SAR Images

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***Abstract*— Ship recognition in synthetic aperture radar (SAR) images has extensive applications across various fields. However, the substantial intraclass variability and interclass similarity present inherent challenges to achieving high-precision recognition. Speckle noise in the background reduces the signal-to-noise ratio, complicating the extraction of discriminative characteristics. In addition, traditional convolutional neural network (CNN)-based methods, which rely solely on image processing frameworks without leveraging additional modality information, struggle to accurately represent the features of SAR targets with bright scatters. To address these issues, we propose a prototype-guided multimodal network (PGMNet), marking a pioneering effort to introduce an image–text multimodal fusion processing paradigm into SAR ship recognition. First, a generation strategy of the prototype and the key area is designed to improve the distinguishability between targets and backgrounds. Besides, a prototype-guided alignment module (PGAM) is implemented to assist the network in characterizing key area information, enhancing intraclass feature consistency. Furthermore, a text feature processing branch is incorporated to precisely describe ship size information and effectively integrate image–text multimodal features, reducing intraclass feature distance while enlarging interclass feature distance. Extensive experiments on the OpenSARShip and FUSARShip datasets demonstrate that the proposed PGMNet achieves state-of-the-art (SOTA) performance. Notably, the accuracy of PGMNet is at least 11% higher than the current SOTA algorithms on the OpenSARShip-VI dataset.**

***Index Terms*— Multimodal feature fusion, prototype guidance, ship recognition, synthetic aperture radar (SAR).**

I. Introduction

ACQUIRING high-resolution remote sensing data has become increasingly accessible due to the continuous advancement of Earth observation technology. In this field, synthetic aperture radar (SAR) offers the advantage of all-weather, all-time observation capabilities and has garnered increasing attention. Automatic target recognition (ATR) is a pivotal task in SAR image interpretation, providing more reliable results and reducing time consumption and financial costs significantly compared to manual annotation. Amongst crucial ATR tasks, SAR ship recognition is currently widely used in port management, maritime route planning, and military surveillance [1], [2], [3], [4], [5].

As deep learning continues to evolve, convolutional neural network (CNN)-based models have become increasingly prevalent in SAR ship recognition. However, the unique imaging mechanism, which depends on the interaction between radar electromagnetic signals and targets, presents numerous challenges for accurate ship recognition in SAR images. As shown in Fig. 1, we select target slices from four types of ships, including bulk carrier, cargo, container ship, and fishing, to better illustrate the characteristics of SAR imaging. On the one hand, a ship may appear significantly different under various conditions, such as resolution, azimuth, polarization modes, and related parameters, leading to huge intraclass diversity. On the other hand, the shape of ships across different subcategories can exhibit significant similarities, causing dramatic interclass similarity. These two factors become intrinsic issues in SAR ship recognition. Currently, it is primarily addressed by enhancing feature extraction capabilities of networks and leveraging additional information [6], [7], [8], [9], [10], [11].

For the former, many existing works applied conventional algorithms from computer vision to SAR ship recognition. Bentes et al. [12] were the first to apply CNN-based models to object classification in oceanographic SAR images, demonstrating the effectiveness of CNN networks for ship recognition in SAR images. Ma et al. [13] proposed a CNN network architecture with six convolutional layers, achieving efficient recognition of ship targets in Gaofen-3 images. Meanwhile, many studies have been focused on the impact of the attention mechanism on ship feature extraction in SAR images. For instance, Wang et al. [6] enhanced the representation capability of multiscale features by utilizing spatial and channel attention, guiding the network to increase the feature difference between categories. Zhao and Lang [7] extracted more discriminative deep transferable features by designing an embedding attention module for subdomain adaptation in SAR ship recognition. Furthermore, many studies have begun exploring feature fusion methods. Li et al. [8] proposed a network with top-down and bottom-up structures to capture long-range dependencies of features, perceive global context information, and enhance ship target representation.

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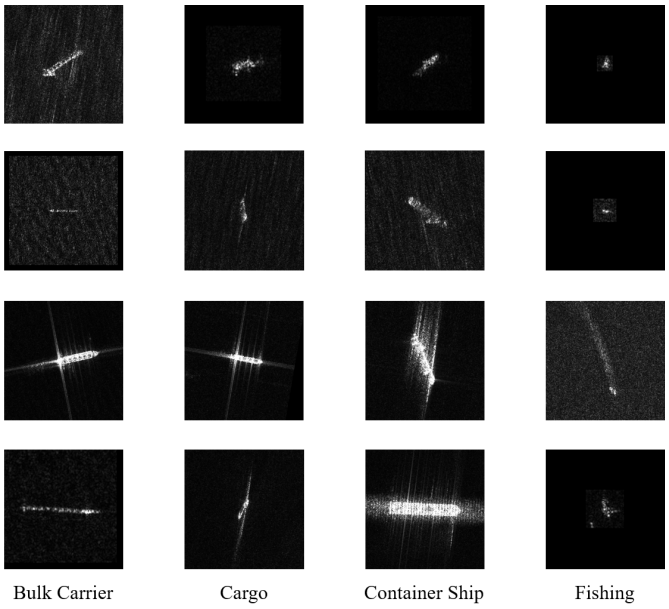


Fig. 1. SAR images of different types of ship targets. From left to right are bulk carriers, cargo ships, container ships, and fishing vessels, totaling four categories.

Wang et al. [9] constructed an adaptive classifier to effectively select and weight multiscale features, increasing intraclass compactness and interclass separability, and achieved good results in SAR ship recognition.

Although these studies have improved feature extraction capabilities by adjusting network structures, they still fail to consider specific challenges. As shown in Fig. 1, speckle noise can degrade the quality of SAR images and reduce the signal-to-noise ratio. The complex background further exacerbates the insufficient interclass differences. Besides, the dihedral-like structures in ships easily form cross-shaped bright spots during imaging, further reducing the feature representation inconsistency within intraclass samples. Consequently, indiscriminately processing the entire image hinders the network's ability to identify crucial discriminative characteristics in SAR ship recognition tasks. To put it another way, the recognition and extraction of the key areas are indispensable.

With the continuous application of multimodal data fusion methods in remote sensing scenarios [14], [15], [16], many studies have begun to utilize additional information in SAR target interpretation to provide complementary support. Inspired by classical methods, Ai et al. [17] effectively improved ship detection accuracy in complex multitarget scenarios by integrating multiscale rotation-invariant features with strong edge characterization capabilities and CNN deep features. Likewise, Zhang et al. [11] combined traditional handcrafted histogram of oriented gradients (HOG) descriptors with CNN-extracted features to achieve precise ship recognition. Meanwhile, many studies have begun to explore the amplitude and phase information stored in the complex format of SAR images. For the task of SAR ship recognition, Zhao et al. [18] developed CVGG-Net, a complex-valued architecture that harnesses both amplitude and phase data from SAR images. This approach outperforms methods that use only real-valued data by incorporating complex-valued area max-pooling for improved feature extraction. At the same time, due to the variations in SAR imaging modes, many studies effectively implement ship recognition using multidimensional remote sensing information from dual-polarized SAR images. For example, Shao et al. [3] incorporated polarization features into their network to improve ship recognition performance. Xie et al. [19] utilized dual-polarization information from SAR images to generate polarimetric feature-guided pseudo-RGB images and texture feature-guided pseudo-RGB images, which more accurately reflect target characteristics for precise ship recognition.

Although the aforementioned methods draw inspiration from multimodal data fusion concepts, they analyze and process SAR ship targets solely from the image's perspective, without incorporating complementary interactions with other modalities, such as text. Moreover, because of the distinct imaging mechanisms of SAR, targets are typically characterized by bright scatterers instead of continuous image forms. Consequently, relying solely on image processing frameworks without adequately leveraging the intrinsic characteristics of SAR images may restrict the expressive capability of SAR data features. This could potentially lead to an induction bias that deviates from human understanding.

Fortunately, even though SAR images are less visually intuitive than optical images, some information in other modalities in SAR and optical images is similar and could be discriminative. Regarding the shapes of certain targets, unlike aircraft that exhibit significant variations under different incident angles in both SAR and optical images, ship targets in SAR images generally retain shape characteristics similar to those seen in optical images. For example, in high-resolution SAR images, a ship's length and width closely match the dimensions of its deck. As a consequence, the geometric information of different ship targets in SAR images still exhibits a high degree of distinguishability (e.g., bulk carriers versus fishing boats), as shown in Fig. 1. This enables us to provide semantic constraints for image interpretation by incorporating text information with prior knowledge, such as the size of the target. This paradigm of image–text fusion offers additional explanation and supplementation in ship recognition and helps reduce the limitations of relying solely on image features.

To address the abovementioned challenges, we propose a prototype-guided multimodal network (PGMNet) for ship recognition in SAR images. Specifically, we employ a prototype and key area generation strategy based on semantic segmentation concepts to extract prototypes containing ships in SAR images. Subsequently, to better suppress the interference of speckle noise, wake, and other clutter in complex backgrounds, we design a prototype-guided alignment module (PGAM) that utilizes the extracted target prototypes to guide the network's feature extraction and representation by focusing on key region information. This enables accurate localization of target areas, allowing the network to concentrate on these high-confidence regions and reducing interference from background noise and irrelevant information. Compared with traditional feature extraction methods that directly apply attention mechanisms, this approach is more targeted and stable.

Meanwhile, the prototype serves as an abstract representation of the target and encapsulates its key semantic information. By leveraging this deep-level semantic information to adjust the image feature extraction process, the output features better align with the target's intrinsic distribution, thereby improving the consistency of intraclass features while increasing the distance between interclass features. Furthermore, to fully leverage target characteristics and effectively utilize multimodal information such as text data for high-confidence ship target classification, we have innovatively designed a text feature processing branch. In this branch, we generate text information containing prior knowledge, such as target size, by utilizing strong scatter points (SSPs) within the target based on SAR imaging mechanisms. We then employ a text information update strategy (TIUS) to map this text information, achieving efficient fusion and representation of image–text multimodal features. Compared with processing based solely on image features, it more effectively leverages the complementarity between different information sources, providing more stable semantic cues that help the network distinguish targets more accurately in cases of high ambiguity or insufficient detail. This paradigm not only alleviates the challenges posed by significant intraclass diversity and interclass similarity but also enhances interpretability, aiding in understanding how the model makes decisions and boosting its transparency and credibility.

Extensive experiments are carried out on published ship recognition datasets, including OpenSARShip and FUSAR-Ship, demonstrating superior performance to other state-of-the-art (SOTA) methods. Notably, on the OpenSARShip-VI dataset, the accuracy of PGMNet is more than 11% higher than the current best CNN-based models, and it converges with significantly fewer training samples. In general, the main contributions of this work can be summarized as follows.

1) Diverging from conventional multimodal fusion in remote sensing, which typically focuses on integrating multisensor visual features, we pioneer the image–text multimodal fusion paradigm in fully supervised SAR target interpretation. Our framework synergistically combines textual semantic embeddings with SAR visual features, significantly enhancing ship recognition performance. To the best of our knowledge, this is the first attempt to introduce an image–text multimodal fusion paradigm into SAR target recognition, providing a new perspective on joint feature representation.
2) To combat interference from speckle noise and background clutter inherent to SAR imaging, we propose the PGAM. This module guides the network to focus on ship-related semantic information by utilizing target prototypes. It effectively suppresses irrelevant signals and enhances feature consistency through structural constraint learning.
3) To fully utilize target characteristics and efficiently integrate text information, we design a text feature processing branch that generates reliable textual descriptors grounded in SAR target attributes and electromagnetic scattering mechanisms. This branch dynamically fuses text and visual features through learnable high-dimensional mappings, enabling physics-constrained feature refinement and complementary information integration.
4) Building on PGAM and our multimodal fusion strategy, we develop PGMNet—a unified network that jointly leverages prototype guidance and image–text fusion to effectively mitigate the challenges of severe intraclass variability and interclass similarity in SAR ship recognition. Extensive experiments on the OpenSARShip and FUSAR-Ship datasets demonstrate that PGMNet attains SOTA performance.

The rest of this article is organized as follows. Section II presents the related work. Section III elaborates on the proposed methodology, while Section IV details experimental results and further analyses. Finally, Section V summarizes the work and provides an outlook for future research.

II. Related Work

In this section, we introduce works related to the proposed method, including the development of SAR ATR, image interpretation guided by key areas, and fine-grained image classification based on image–text multimodal features.

A. ATR of SAR

Depending on whether the features need to be constructed manually, current research for SAR ATR can be broadly categorized into two approaches: traditional methods and deep learning methods.

Template-based methods [20], [21], [22], [23], [24] and model-based methods [25], [26], [27] are the two primary strategies for traditional SAR ATR. The template-based method mainly adopts the approach of "feature extraction and classifier." Its key lies in feature extraction and selection, which requires profound domain knowledge as the basis. Typical features include geometric, texture, electromagnetic scattering, and transform domain features. This method is intuitive and rudimentary. However, it is difficult to ensure the robustness of features. The core of the model-based approach lies in the design of the model. By the physical modeling of the target, the scattering process of electromagnetic wave irradiation on the target is simulated, and the features of the target are predicted under different attitudes, configurations, and environmental conditions. However, the target's parameter information is required in advance. Besides, as resolution improves, the computation burden of physical modeling and the difficulty of simulation increase accordingly, making it impossible to meet the demands of efficiency in practical applications.

In recent years, deep learning has been applied to SAR target interpretation due to its robust automatic feature extraction capability. Unlike traditional methods, deep learning technology automatically identifies effective features within data. This ability often results in better recognition performance than models requiring handcrafted features. Shao et al. [28] evaluated different CNN models, including AlexNet, VGGNet, and so on, on the MSTAR dataset, analyzing accuracy, computing resources, and gradient issues, providing a reference for CNN applications in SAR ATR. Chen et al. [29] designed a fully

convolutional network to achieve high-precision identification of terrestrial SAR image targets. In addition, studies have begun to notice the differential impact of various feature representations on classification performance, with an increasing number of attention mechanisms applied to SAR ATR. For instance, Wang et al. [30] applied SENet [31] to extract effective features and obtain good performance in the MSTAR dataset. Meanwhile, numerous studies [32], [33], [34], [35], [36] have applied other advanced attention modules such as CBAM [37] to their target recognition networks for SAR images.

Although these networks improved the performance compared to traditional methods, they still lack the specific integration of attention mechanisms with the characteristics of SAR images. Therefore, in the task of SAR ship recognition, we combine the inherent imaging characteristics of SAR images, starting from the perspective of semantic segmentation, and isolate target prototypes that contain key information. This process separates them from backgrounds with speckle noise, thereby enhancing the representational capacity of discriminative features.

At the same time, in addition to visual features, existing studies utilize supplementary information to enhance recognition accuracy. Some work [38], [39] exploits the unique complex form of SAR images, using phase information to enhance feature representation. Li et al. [40] have combined global features extracted by CNNs with structural features from complex data, known as attributed scattering center (ASC) components, to construct a more robust and efficient classification network. Zeng et al. [41] developed an architecture, MS-CVNets, for complex-valued SAR target recognition. It utilizes specific operations and structures to consider phase information, with complex-valued building blocks and feature fusion strategies, achieving better performance than real-valued methods on the MSTAR dataset. Others [19], [42] have designed frameworks for processing SAR images with multiple polarization modes, fully leveraging the different responses of materials to various polarizations to maximize the utility of image information.

Despite these methods achieving promising results, significant challenges persist in SAR ship recognition, particularly with single-polarization SAR images. This limitation arises from relying solely on image modality processing frameworks, which may fail to accurately distinguish between similar interclass targets. Therefore, to maximize the model's ability to learn small differences between ships from different classes, in this article, we innovatively propose a text–image multimodal feature fusion processing paradigm, exploring the benefits of multimodal features in SAR ATR.

B. Image Interpretation Guided by Key Areas

In CNN-based image interpretation, different pixels contribute differently to the results, and simply treating all pixels equally with convolution kernels often fails to capture discriminative features adequately. Most methods enhance the activation of key pixels by introducing attention mechanisms while suppressing background noise. Also, many studies have started directly utilizing key region guidance to extract and represent image features. In medical image processing, Doan et al. [43] proposed an architecture leveraging ordered regression map self-guidance for nuclei segmentation, which adaptively adjusts the extraction weights for different pixels. Li et al. [44] incorporated full lung information in diagnosing lung cancer with the guidance of masks, which learned global and local features effectively and improved the diagnostic accuracy of tumors. At the same time, the concept of key area guidance has also been applied to enhance image interpretation capabilities in remote sensing image processing. For example, Hou et al. [45] utilized a mask-guided mechanism in image registration, assisting the network training process through the input thresholding mask and output transformation mask to solve the problem of misalignments. Meanwhile, Wang et al. [46] employed distinct feature extraction branches to capture target, background, and overall features. They introduced a guidance mechanism leveraging structural information to enhance feature representation, achieving SOTA performance in the scene classification task of remote sensing images.

Inspired by these works, this article combines the imaging characteristics of SAR images and proposes a prototype-guided alignment structure. With the help of semantic segmentation concepts, the key areas representing ships are obtained. The network is guided to stimulate the representation of ship features and suppress background interference containing speckle noise, effectively improving the consistency of intraclass features.

C. Multimodal Fusion in Remote Sensing Image Interpretation

In recent years, multimodal fusion technology has made significant progress in computer vision [47], [48], [49], and an increasing number of studies have sought to apply this paradigm to target detection and recognition in remote sensing images. Representative approaches include fusing optical and infrared images for enhanced detection accuracy [50], [51], integrating optical imagery with corresponding height maps to improve vehicle detection in complex scenarios [52], and combining optical data with LiDAR information for 3-D object detection [53], [54]. In SAR image interpretation, existing multimodal applications primarily focus on combining image features with scattering characteristics and phase information to enhance target perception [55]. It is clear that current multimodal fusion approaches for remote sensing target detection and recognition remain primarily sensor-dependent, focusing on data integration from disparate imaging sources. This paradigm, however, fails to explore the synergistic potential of nonsensor-based representations such as text that could complement conventional fusion frameworks. As a result, this limitation remains a fundamental constraint of image-centric processing, especially when dealing with SAR images contaminated by speckle noise.

Regarding image–text fusion paradigms, current remote sensing research mainly explores three directions: zero-shot learning [56], image captioning [57], and image retrieval [58].

Nevertheless, the absence of effective and reliable textual annotations has become a critical bottleneck for implementing this paradigm in fully supervised target detection and recognition. In this article, we propose a novel solution rooted in SAR imaging mechanisms and emphasizing target characteristics. We exploit discriminative prior knowledge—including target length and width—as text information for multimodal fusion. This strategic integration significantly strengthens the coupling between visual and textual modalities, thereby effectively improving ship target recognition performance in SAR imagery.

III. Methodology

This section provides an overview of the proposed PGMNet in Section III-A. The prototype and key area generation strategy are detailed in Section III-B. Subsequently, the proposed PGAM in the image feature processing section and the text feature processing branch are introduced in the following two parts.

A. Overall Framework

The overall structure of the proposed PGMNet network is illustrated in Fig. 2. First, the idea of semantic segmentation is adopted to obtain the corresponding target prototype from the input original SAR ship images. Then, the key area containing ship target information is generated by multiplying the target prototype by the original image. Subsequently, multimodal features are processed through the image feature processing branch and text feature processing branch.

For the image processing branch, the original SAR ship image and the corresponding target prototype are used as inputs. The image features from the original image are extracted through various feature extraction stages of the backbone network. The designed PGAM aligns the features, from the second and fourth layers, with the resized target prototype. The earlier aligned feature layer is then fused with the final aligned feature layer through a residual structure and a concatenation operation. In this way, the shallow features rich in edge details are integrated with the deep features that capture broader structural information. Specifically, in this work, ResNet50 [59] is utilized as the backbone network.

For the text feature processing branch, the input is the key area obtained from the previous step, which contains ship target information. First, the SSPs are extracted and fit to derive text information describing the ship target. As shown at the bottom of Fig. 2, the red dots represent the extracted SSPs, while the green lines represent the fit length and width of the ship target. Then, the designed TIUS updates the text information into the image features through adaptive projection. This realizes information interaction in higher and wider dimensions. It also increases the distinction between different types of ships and improves the network recognition ability.

The proposed PGMNet not only uses the prototype to explore the characteristics of the image fully but also supplements the inherent text information features of the ship to enrich the feature expression form. This effectively shortens the feature distance within the intraclass and clarifies the differences in object attributes between classes.

B. Prototype and Key Area Generation Strategy

Generally, an SAR ship slice consists of key area information about the target and a low information frequency background mixed with speckle noise. Since ship recognition primarily relies on discriminative class-specific features contained in these critical areas of targets, indiscriminately processing the entire input sample proves suboptimal. In this article, we employ the concept of semantic segmentation to extract target prototypes and key areas that potentially map to ship targets.

The processing pipeline begins with applying the classical Lee filter [60] to suppress speckle noise interference in the background. Subsequently, a segmentation method based on adaptive thresholds is employed to select areas containing crucial information about the ship. This selected area includes pixels characterizing the target and portions of the coherent noise in the SAR image background. Considering that the geometric representation of ship targets in SAR images may be discontinuous, the morphological closure method is applied to connect various potential areas, and the largest connected area is selected. Then, morphological opening operations are performed to eliminate clutter further, resulting in the final ship prototype. This process effectively suppresses background noise interference while maintaining the geometric integrity of targets. Finally, the key area representing the ship's characteristics is obtained by multiplying the target prototype by the input image.

C. Prototype-Guided Alignment Module

In this section, PGAM is designed to process image features in PGMNet. This module aligns image features with target prototypes, enhancing intraclass consistency and mitigating performance degradation caused by large intraclass feature variations.

Specifically, as shown in Fig. 3, let the input image feature layer be denoted as $I \in \mathbb{R}^{c\times h\times w}$, where c, h, and w represent the channels, height, and width of the image feature layer, respectively. First, the extracted ship prototype is unified to the same size, denoted as $M \in \mathbb{R}^{1\times h\times w}$. Then, we generate the corresponding weight matrices using learnable kernels and adjust their sizes accordingly. These detailed processes are shown in the following equations:

$$W_v = R[\Psi_v(I)] \tag{1}$$

$$W_q = R[\Psi_q(I)] \tag{2}$$

$$W_k = R[\Psi_k(M)] \tag{3}$$

where $R(\cdot)$ indicates the reshape operation, and $\Psi_v(\cdot)$, $\Psi_q(\cdot)$, and $\Psi_k(\cdot)$ denote the learnable kernels. W_v, W_q, and W_k are all unified to $\mathbb{R}^{c\times hw}$.

Notably, unlike traditional self-attention mechanisms, the weight matrices in the proposed PGAM are derived separately from the image feature layers and the corresponding ship prototype. This further models rich relationships over local

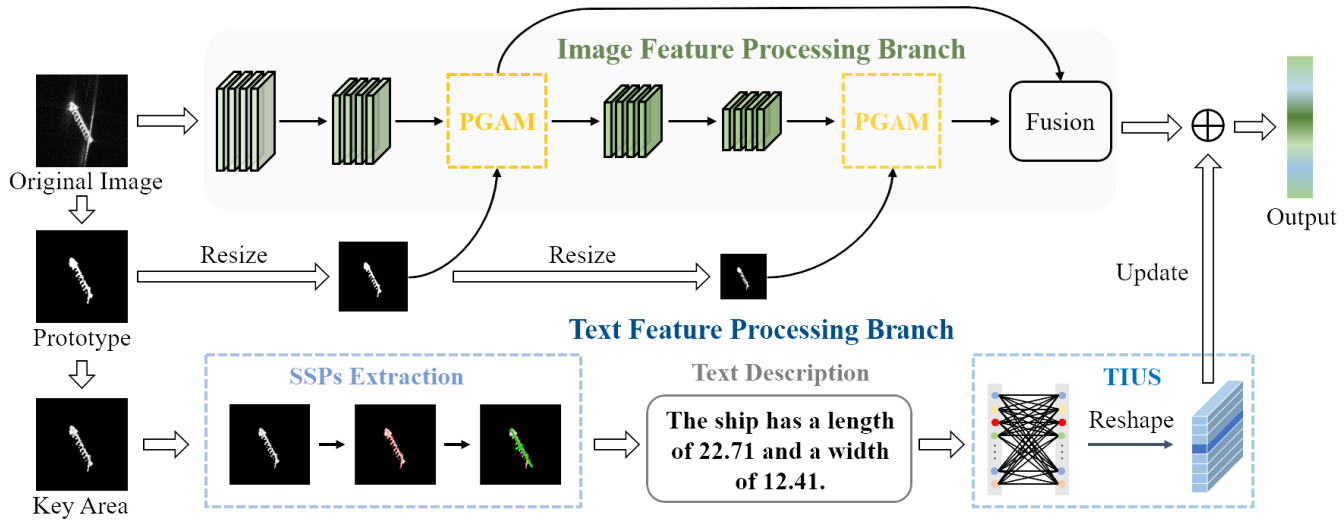


Fig. 2. Overall structure of the proposed PGMNet. The flowchart illustrates the PGMNet framework, where the left part depicts the generation process of target prototypes and key regions. Subsequently, the original image and corresponding prototype are inputted to extract visual features through the image feature processing branch, which includes the PGAM. Simultaneously, in the text feature processing branch, strong scattering points (SPPs) are extracted from key areas to generate reliable text descriptions. The TIUS is then employed to efficiently integrate text–image features, thereby accomplishing the SAR ship recognition task.

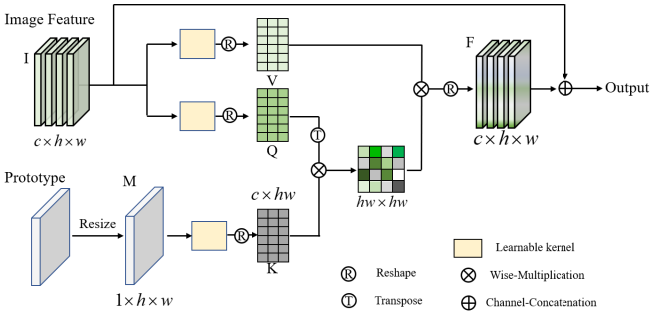


Fig. 3. Structure of the proposed PGAM. This module utilizes prototypes corresponding to image features to guide the network's attention toward critical regions, thereby enhancing the consistency of feature representation.

features in key areas and aligns the image features with the ship prototype for representation.

Guided by the prototypes, broader contextual information can be encoded into local features of interest, enhancing the inherent feature representation of the class. This improves intraclass compactness and semantic consistency. Specifically, we achieve alignments through the following equation:

$$F = R\left[W_V \cdot \text{softmax}\left(\frac{W_Q^{\text{T}} \cdot W_K}{\sqrt{d_k}}\right)\right] \tag{4}$$

where F represents the features obtained after aggregation, softmax$(\cdot)$ indicates the operation using the softmax activation function, T$(\cdot)$ denotes the transpose operation, and d_k is a normalization factor.

Finally, the output of the PGAM is the concatenation of the input image feature I and the aligned feature F

$$\text{Output} = \text{Concat}(I, F) \tag{5}$$

where Concat$(\cdot)$ represents the concatenation of channels.

By aligning image features with target prototypes, the proposed PGAM enhances discriminative feature representation

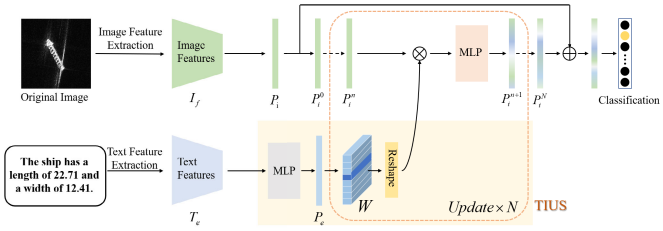


Fig. 4. Structure of the proposed text feature processing branch. This branch encodes text descriptions into text features and iteratively and adaptively fuses them with previously processed image features using the TIUS.

capabilities. This approach optimizes the network's capacity to generate compact and consistent feature embeddings in key regions. Simultaneously, it actively suppresses interference from background noise patterns. These coordinated mechanisms collectively strengthen the inherent feature representation of the class.

D. Text Feature Processing Branch

This part introduces the generation method for text descriptions and the TIUS designed to achieve image–text alignment and fusion. As shown in Fig. 2, we first fit the strong scatterer points (SSPs) in the key area of ship images to summarize and extract text descriptions of target characteristics. These text descriptions are then encoded using a text feature encoder. Subsequently, TIUS is applied to dynamically and adaptively project the intrinsic morphological features of the target, enabling multimodal alignment between text and visual features. This synergistic integration ensures a comprehensive fusion of complementary information, thereby enhancing ship recognition accuracy.

1) Text Description Generation: SAR images are susceptible to speckle noise interference, often resulting in degraded visual details. To alleviate this limitation, text information

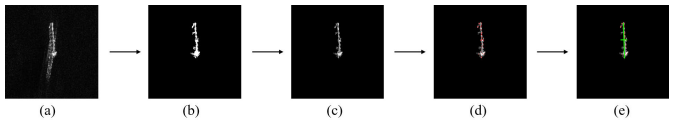


Fig. 5. Extraction process of SSPs. (a)–(e) Original SAR image, the prototype corresponding to the ship target, the key area containing the ship target, the extracted SSPs, and the obtained ship dimension information, respectively.

is innovatively employed to supplement semantically critical features absent in the imagery. Unlike previous studies [61], [62] that generate text descriptions using large-scale models (e.g., CLIP or ChatGPT), our approach is founded on the SAR imaging mechanism. It exploits the scattering characteristics of the target to extract prior knowledge relevant to its properties. Compared with text generated from models pretrained on natural scenes, our description is more reliable for interpreting SAR imagery.

We incorporate key attributes—including target length and width—into the text descriptions and standardize the text description format: The ship has a length of L and a width of W, as shown in Fig. 4. This approach imposes semantic constraints that mitigate significant intraclass variations while enlarging interclass distance.

It is noteworthy that radar waves exhibit strong backscattering at the ship's structures with dihedral and trihedral corners, such as between the containers of cargo ships, near the oil pipelines of tankers, and around the bridge, bow, and stern of the ships. In SAR images, this backscattering appears as bright areas, and SSPs remain relatively stable under varying external conditions. Therefore, we utilize SSPs to describe ship targets in SAR images in this work. To extract SSPs from key areas of target slices, we employ the Harris–Laplace detector, which is known for its strong invariance to rotation and its robustness in feature detection [63]. The SSP extraction process is illustrated in Fig. 5(a)–(d).

Specifically, on account of the axial symmetry of the ship, the least-squares method is adopted to carry out linear regression for the SSPs to extract the straight line of the ship. The specific calculations for the least-squares method are given as follows: assume that the extracted SSPs are expressed as $\left[(x_i, y_i), i \in \{1, 2, \ldots, n\}\right]$, and the center of the sample points $(\overline{x}, \overline{y})$ is calculated by

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i. \tag{6}$$

Then, we can calculate the intercept and the slope of the line to get the line through the ship's hull

$$b = \frac{\sum_{i=1}^{n} x_i y_i - n\overline{x}\,\overline{y}}{\sum_{i=1}^{n} x_i^2 - n\overline{x}^2} \tag{7}$$

$$k = \overline{y} - b\overline{x}. \tag{8}$$

The length of the hull can be represented by the distance between the two points of maximum separation on the line. Then, the longest segment in the vertical direction of the hull is extracted. These two lengths represent the ship's length and width, respectively, as shown in Fig. 5(e).

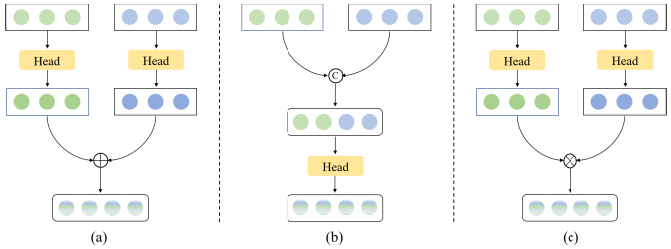


Fig. 6. Traditional methods for fusing image and text information. (a) Addition. (b) Concatenation. (c) Multiplication.

2) Text Information Update Strategy: Previous works [64], [65], [66], [67], [68] typically fuse image features and text information directly through operations such as addition, concatenation, or multiplication before the final classification head, as illustrated in Fig. 6(a)–(c). However, the inherent semantic gap between image and text features means that such straightforward fusion does not yield optimal results. In this article, as depicted in Fig. 4, we propose a new fusion strategy, TIUS. This approach maps the text modality into a high-dimensional space and iteratively fuses it with the image features adaptively. This effectively bridges the cross-modal semantic disparity between images and texts, facilitating a more comprehensive integration of the relevant information.

Specifically, as illustrated in Fig. 4, the original SAR image undergoes the previously described image processing structures to extract high-level features. Then, we adopt global average pooling (GAP) to compress spatial dimensions, yielding a compact image feature vector $P_i \in \mathbb{R}^{B\times C}$, where B is the batch size and C denotes the image channel dimension. For the input text information, we employ a CLIP-pretrained Transformer model [69] to map it into text features $T_e \in \mathbb{R}^{B\times N}$, where N is the text feature dimension. CLIP, trained via contrastive learning, ensures semantic alignment between text and image features in a shared latent space, providing a robust foundation for cross-modal interaction. Subsequently, the text features T_e are projected to a broader channel dimension via a multilayer perceptron (MLP) to obtain P_e. This projection facilitates the transformation of text features into a high-dimensional space. As a result, the expressiveness of the text information is significantly enhanced. It is expressed by the following formula:

$$P_e = \mathrm{MLP}(T_e) = \mathrm{ReLU}(\mathrm{LN}(f(T_e))) \tag{9}$$

where $P_e \in \mathbb{R}^{B\times K}$, $K = C \times C$, $\mathrm{ReLU}(\cdot)$ stands for ReLU activation function, $\mathrm{LN}(\cdot)$ stands for the layer normalization, and $f(\cdot)$ represents the fully connected layer.

Next, we obtain the high-dimensional dynamic modulation weight W by employing a fully connected layer and a reshape operation as

$$W = \mathrm{Reshape}(f(P_e)) \tag{10}$$

where $W \in \mathbb{R}^{B\times C\times C}$ and $\mathrm{Reshape}(\cdot)$ denotes that 2-D features are reconstructed into 3-D features. These weights adaptively encode text semantics to guide the refinement of image features.

Finally, at each iteration, the image features P_i^n are modulated by the text-guided weights W via matrix multiplication, producing refined features. This operation inherently acts as a text-driven attention mechanism, selectively enhancing or suppressing regions of the image features based on text semantics. For instance, regions correlated with text descriptors such as "ship size" are accentuated. An MLP further processes the modulated features to obtain features P_i^{n+1} to enhance nonlinear representational capacity. The detailed processes are shown in the following equation:

$$P_i^{n+1} = \mathrm{MLP}\big(\mathrm{mul}\big(P_i^n; W\big)\big) \tag{11}$$

where P_i^n represents an image dynamically updated n times, wherein $n \in \{1, 2, \ldots, N\}$, and $\mathrm{mul}(\cdot)$ represents multiplication operation. The above procedure uses P_i^0 and P_e as inputs and is iterated n times until it completes the Nth recursion. In this work, N is empirically set as 3. This iterative updating strategy enables textual cues to refine visual features, gradually emphasizing discriminative patterns.

IV. Experiments and Results

This section presents extensive experiments and results to highlight the excellent performance of the proposed PGMNet in SAR ship recognition. We begin by detailing the datasets used in this work, along with our experimental setup and evaluation metrics. Subsequently, we compare PGMNet against other SOTA methods to demonstrate its superiority and include ablation studies to validate the effectiveness of the designed modules. Finally, we analyze the impact of different ship size extraction methods, varying training samples, and various multimodal information fusion methods on network performance.

A. Datasets

1) OpenSARShip Dataset: OpenSARShip, proposed by Huang et al. [70] in 2018, includes 11 346 ship slices. These slices are sourced from 41 Sentinel-1 satellite images covering five ports: Tianjin Port, Shenzhen Port, Shanghai Port, Yokohama Port, and Singapore Port. The dataset provides detailed types from AIS information. The ship slices are captured in IW mode with VV–VH polarization, exhibiting various ship lengths, directions, and incidence angles.

In this work, we exploit single look complex (SLC) products that have a range resolution of 2.7–3.5 m and an azimuth resolution of 22 m. Similar to previous works [11], [71], [72], we conduct extensive experimental analysis on two datasets filtered by elaborated types to evaluate the proposed PGMNet. One dataset includes three categories of ships: tanker, bulk carrier, and container ship, referred to as the OpenSARShip-III dataset in this article. The more challenging dataset contains six categories, general cargo, cargo, tanker, container ship, and fishing, and is referred to as the OpenSARShip-VI dataset in this article. The details of the two datasets are shown in Tables I and II. In addition, we list the average length and width of ships for different categories in the tables. Fig. 7 displays sample slices of ships from the OpenSARShip dataset.

TABLE I
Partition Between the Training and Test Sets on the OpenSARShip-III Dataset. Besides, the Last Two Columns Depict the Length and Width of the Targets in Each Category

Dataset	Category	Training	Test	All	Length	Width
OpenSARShip-III	Bulk Carrier	338	328	666	216	34
	Container Ship	338	808	1146	255	35
	Tanker	338	146	484	101	16

TABLE II
Partition Between the Training and Test Sets on the OpenSARShip-VI Dataset. Besides, the Last Two Columns Depict the Length and Width of the Targets in Each Category

Dataset	Category	Training	Test	Alll	Length	Width
OpenSARShip-VI	Bulk Carrier	200	466	666	216	34
	Cargo	200	1142	1342	128	20
	Container Ship	200	946	1146	255	35
	Fishing	200	50	250	19	5
	General Cargo	200	84	284	124	20
	Tanker	200	284	484	101	16

2) FUSAR-Ship Dataset: FUSAR-Ship, introduced by Hou et al. [73] in 2019, includes over 5000 ship slices derived from 126 Gaofen-3 satellite images operating in C-band. Images on the FUSAR-Ship dataset have higher resolution compared to the OpenSARShip dataset, with an azimuth resolution of 1.124 m and a range resolution greater than 2 m. Similar to previous works [10], [11], [71], we select seven categories of ship targets for our experiments: general cargo, container ship, other cargo, bulk carrier, tanker, fishing, and other. Table III presents the sample count, along with the average length and width for each category in the experiments. Fig. 8 displays some sample slices of the ships.

B. Experiment Details

1) Implementation Setting: All experiments are carried out and analyzed on OpenSARShip-III, OpenSARShip-VI, and FUSAR-Ship datasets. During the experiments, all images are unified to 224 × 224 pixels. PGMNet is trained for 100 epochs, and the batch size is set to 64. All experiments are implemented by PyTorch 1.7.1 on an NVIDIA RTX A6000. Meanwhile, we select stochastic gradient descent (SGD) as the optimizer, with an initial learning rate of 0.01 and a momentum term of 0.9 to suppress oscillations during optimization.

2) Loss Function: In line with comparative methods [11], [19], the cross-entropy loss function is adopted in this work to optimize the proposed network.

3) Evaluation Metrics: In this article, similar to [10], [11], [71], we primarily employ accuracy as the main metric to evaluate network performance. Furthermore, we provide additional metrics such as precision, recall, and $F1$-score to comprehensively reflect the network's performance.

C. Comparison With CNN-Based Models

To fully validate the effectiveness of the proposed PGMNet for SAR ship recognition, we select several representative CNN-based networks for comparison. Among these methods, AlexNet [74], VGG-16 [75], ResNet-50 [59], DenseNet-161 [76], MobileNet-V2 [77], SqueezeNet [78], InceptionV3 [79],

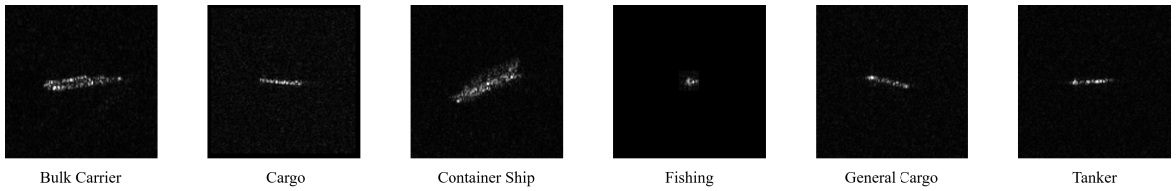


Fig. 7. SAR ship slices in the OpenSARShip Dataset.

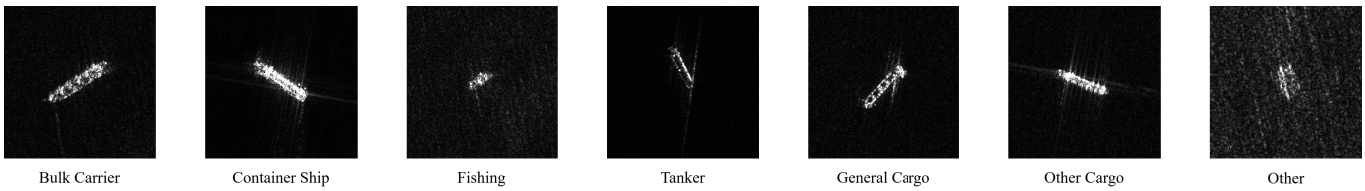


Fig. 8. SAR ship slices in the FUSAR-Ship Dataset.

TABLE III
PARTITION BETWEEN THE TRAINING AND TEST SETS ON THE FUSAR-SHIP DATASET. BESIDES, THE LAST TWO COLUMNS DEPICT THE LENGTH AND WIDTH OF THE TARGETS IN EACH CATEGORY

Dataset	Category	Training	Test	All	Length	Width
FUSAR-Ship	Bulk Carrier	1150	494	1644	165	24
	Container Ship	1219	523	1742	213	30
	Fishing	1101	473	1574	35	6
	Tanker	1215	521	1736	84	14
	General Cargo	1205	517	1722	122	17
	Other Cargo	1214	521	1735	77	14
	Other	1211	520	1731	54	10

InceptionV4 [80], and Xception [81] are classical classification networks that have shown good performance in natural scenes. In addition, we select some of the latest SAR ship recognition methods. For instance, Wang et al. [82] combine pretrained model weights with CNN models to enhance SAR ship recognition performance. Lyu et al. [83] utilize a mix attention mechanism to improve the network's representation of target region features. Zeng et al. [84] and SE-LPN-DPFF [72] use VV–VH dual-polarization SAR images as input for dual-polarization image processing networks. Furthermore, we consider networks that utilize additional information, such as MFCFNet [10], which introduces handcrafted features, HOG-ShipCLSNet [11], which incorporates HOG features, and MFJL [85], which combines pattern features, to explore the impact of supplementary information for feature representation. MetaBoost [86], an integrated approach using a two-stage filtering process to select the most diverse and accurate classifiers from a pool of base classifiers, demonstrates strong robustness in SAR ship recognition. HDSSNet [71], a method that specifically addresses the challenge of large scale variations in ships by designing a dedicated feature extraction module for SAR imagery, thereby enhancing discriminative feature extraction.

To demonstrate the performance of the aforementioned methods under different scenarios, OpenSARShip-III and OpenSARShip-VI datasets and the FUSAR-Ship datasets are employed for performance comparison and analysis. The results are shown in Table IV. To ensure the fairness of the experiments, we utilize publicly available data for some non-open-source methods, selecting the corresponding dataset metrics as the basis for comparison. Notably, the methods by Zeng et al. [84] and SE-LPN-DPFF [72] require dual-polarization SAR data as input, making them unsuitable for the seven-category FUSAR-Ship dataset. In addition, since we could not find publicly available data for Lyu et al. [83], MFCFNet [10], MFJL [85], and MetaBoost [86] on the OpenSARShip-VI dataset, they are not discussed here. Besides, the best results are highlighted in bold, and the second-best results are underlined.

1) Recognition Results on the OpenSARShip-III Dataset: In the OpenSARShip-III dataset, the proposed PGMNet achieves the highest accuracy of 88.07%, which is more than 10% higher than classical CNN models and at least 4% higher than other SOTA SAR ship recognition models, demonstrating superior performance. Among the remaining methods, HDSSNet [71] achieves the second-best result with an accuracy of 83.23%. Other top-performing networks are MetaBoost [86], SE-LPN-DPFF [72], and MFJL [85], with accuracies being 81.01%, 79.25%, and 79.02%, respectively. Among the classical CNN networks, VGG-16 offers the highest performance, achieving an accuracy of 76.76%.

2) Recognition Results on the OpenSARShip-VI Dataset: The OpenSARShip-VI dataset includes three additional categories: cargo, general cargo, and fishing, which further increase the similarity between different ship types and present significant challenges for recognition networks. As shown in Table IV, the networks specifically designed for SAR ship recognition outperform traditional CNN networks. Apart from PGMNet, the top four networks are HDSSNet [71], SE-LPN-DPFF [72], Zeng et al. [84], and HOG-ShipCLSNet [11], with accuracies of 60.76%, 56.66%, 55.26%, and 53.77%, respectively. Most networks involved in the comparison achieve an accuracy of less than 60%. However, PGMNet achieves an accuracy of 72.34%, which is 11.58% higher than the suboptimal method, HDSSNet [71], significantly improving recognition performance. This proves that PGMNet excels in various scenarios with small interclass differences.

3) Recognition Results on the FUSAR-Ship Dataset: The FUSAR-Ship dataset provides higher resolution compared to

TABLE IV
QUANTITATIVE COMPARISON BASED ON CNN-BASED MODELS. NOTE THAT THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. AND THE SECOND-BEST RESULTS ARE UNDERLINED

Model	OpenSARShip-III				OpenSARShip-VI				FUSAR-Ship			
	Acc(%)↑	R(%)↑	P(%)↑	F1(%)↑	Acc(%)↑	R(%)↑	P(%)↑	F1(%)↑	Acc(%)↑	R(%)↑	P(%)↑	F1(%)↑
AlexNet	72.78	67.12	70.69	67.92	40.38	37.45	52.33	34.14	73.94	75.03	73.75	73.85
VGG-16	76.76	72.28	78.03	74.65	48.76	44.05	58.83	42.36	77.19	78.47	76.97	77.1
ResNet-50	73.48	67.03	76.67	69.9	47.71	38.52	48.53	36.64	81.68	82.39	81.38	81.36
DenseNet-161	75.66	70.39	73.87	70.43	52.49	41.75	54.35	42.09	82.15	82.57	81.95	81.93
MobileNet-v2	66.07	60.42	64.19	61.88	46.9	39.39	52.19	38.48	79.46	80.22	79.31	79.27
SqueezeNet-v1	73.56	69.64	76.75	72.33	49.23	40.35	46.26	38.52	72.09	73.78	71.95	72.14
Inception-v3	72.62	66.82	76.57	69.83	44.21	37.24	44.3	30.77	79.13	79.89	78.88	78.81
Inceptionnn-v4	68.02	63.29	74.68	65.74	44.78	37.92	51.72	36.2	80.39	81.2	80.15	80.13
Xceptionn	70.75	64.6	75.15	66.98	51.11	41.98	53.34	41.56	83.16	84.06	82.95	82.9
Wang et al.	69.27	57.72	58.72	58.12	50.53	50.53	41.77	45.73	80.02	80.09	80.32	80.21
Lyu et al.	77.54	73.61	73.07	73.23	-	-	-	-	-	-	-	-
Zeng et al.	77.41	74.99	74.05	74.52	55.26	55.66	47.16	50.96	-	-	-	-
SE-LPN-DPFF	79.25	78.83	76.45	77.62	56.66	54.49	48.61	51.38	-	-	-	-
MFCFNet	78.6	79.37	73.78	76.06	-	-	-	-	87.23	86.67	86.89	86.69
HOG-ShipCLSNet	78.15	77.87	72.42	75.04	53.77	54.2	46.66	50.15	86.69	86.62	86.54	86.58
MFJL	79.02	73.97	78.38	76.11	-	-	-	-	87.59	87.86	88.54	88.20
MetaBoost	81.01	80.9	76.64	78.17	-	-	-	-	90.99	90.83	90.9	90.85
HDSSNet	83.23	80.66	81.16	80.91	60.76	59.53	60.26	59.89	90.08	90.08	90.15	90.12
PGMNet(Ours)	**88.07**	**87.87**	**89.03**	**88.44**	**72.34**	**64.26**	**72.53**	**66.7**	**93.19**	**93.05**	**93.10**	**93.05**

the OpenSARShip dataset. Therefore, as shown in Table IV, the recognition performance on this dataset is generally higher. On this dataset, PGMNet achieves the highest accuracy of 93.19%, surpassing the other networks in the top five performers—MetaBoost [86], HDSSNet [71], MFJL [85], and MFCFNet [10]—by 2.2%, 3.11%, 5.60%, and 5.96%, respectively. This demonstrates its robustness and extraordinary recognition performance across different resolutions and scenarios.

4) Recognition Performance of Classical CNN-Based Models: These classical methods achieved accuracies of no more than 80%, 53%, and 84% on the OpenSARShip-III, OpenSARShip-VI, and FUSAR-Ship datasets, respectively. This is mainly because classical CNN-based methods are primarily designed for natural scenes and do not account for the special imaging mechanisms of SAR. Specifically, SAR images differ from optical images, including speckle noise in the background of ship targets and cross-shaped bright spots in dihedral-like structures of ships. Direct application often fails to yield good results. Among the classical CNN-based algorithms, the Xception network offers relatively good performance. Introducing depthwise separable convolutions improves efficiency in processing SAR images compared to traditional convolutions. However, it still underperforms compared to the latest methods specifically designed for SAR ship recognition.

5) Recognition Performance of Novel SAR Ship Recognition Methods: The novel SAR ship recognition methods selected for this work generally outperform classical CNN-based methods. Specifically, Zeng et al. [84] and SE-LPN-DPFF [72] are dual-polarization SAR ship recognition networks. SE-LPN-DPFF [72], with its squeeze-and-excitation Laplacian pyramid structure, handles input VV–VH dual-polarization images more effectively, achieving better results. However, it still achieves less than 80% accuracy on the OpenSARShip-III dataset and less than 60% on the OpenSARShip-VI dataset, indicating room for improvement. Networks utilizing additional information (e.g., HOG-ShipCLSNet [11], MFCFNet [10], and MFJL [85]) have demonstrated improvements compared to single-feature networks, yet they still exhibit limitations. HOG-ShipCLSNet [11] integrates HOG features with image features, but the quality of its handcrafted HOG features—particularly in low-resolution OpenSARShip datasets—often leads to suboptimal recognition performance. Similarly, MFCFNet [10] incorporates various handcrafted features, and MFJL [85] integrates pattern features, but both face analogous challenges in feature quality and fusion effectiveness. Furthermore, while leveraging auxiliary information, these networks fail to transcend the constraints of traditional image processing frameworks, remaining confined to the integration and utilization of visual features. In contrast, PGMNet effectively extracts text information describing SAR ship size and combines it with image features to achieve high-accuracy recognition. MetaBoost [86] and HDSSNet [71] are methods with suboptimal performance. MetaBoost [86] integrates many base classifiers and selects different combinations for different datasets, but this approach is relatively complex and has high training costs. HDSSNet [71] designs a specialized feature extraction module for SAR ship images and a spherical space classifier. It performs suboptimally on the OpenSARShip dataset but still falls short by 11% in accuracy compared to PGMNet on the OpenSARShip-VI dataset, thus demonstrating the limitations of using only image processing frameworks for SAR ship recognition tasks.

6) Recognition Performance of PGMNet: As shown in Table IV, PGMNet achieves the highest accuracy in all three datasets, with 88.07% on the OpenSARShip-III dataset, 72.34% on the OpenSARShip-VI dataset, and 93.19% on the FUSAR-Ship dataset. PGMNet also scores highest in other metrics, including recall, precision, and $F1$-score. Notably, on the OpenSARShip-VI dataset, PGMNet outperforms the suboptimal network by over 11% in accuracy. These results demonstrate that PGMNet delivers excellent performance across different datasets, resolutions, and scenarios for ship

TABLE V
ABLATION EXPERIMENTS FOR PGMNET. NOTE THAT THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Model	PGAM	Text Feature Processing	Accuracy(%)↑	
			OpenSARShip-III	OpenSARShip-VI
Baseline	✗	✗	73.48	47.71
Baseline-Prototype	✓	✗	80.73	54.51
Baseline-Text	✗	✓	83.46	67.87
PGMNet	✓	✓	**88.07**	**72.34**

recognition in various SAR images. The proposed image–text multimodal information fusion paradigm, guided by target prototypes, enhances interclass diversity and intraclass feature consistency. This results in better learning of discriminative features and improves SAR ship recognition performance.

D. Ablation Study

To better validate the effectiveness of the modules designed in PGMNet, we conduct a comprehensive experimental analysis on the OpenSARShip-III, OpenSARShip-VI, and FUSAR-Ship datasets. In these experiments, ResNet50 is selected as the baseline. The method baseline-prototype means that the PGAM is added to the baseline, but the corresponding text features are not processed. Baseline-text means that multimodal text features are added to the network architecture, but ResNet50 is still used as the backbone network for image processing without using the PGAM.

1) Quantitative Analysis: The results of experiments conducted on different networks using the OpenSARShip datasets are presented in Table V.

The table demonstrates that after adding the PGAM, the accuracy on the OpenSARShip-III dataset increases from 73.48% to 80.73%, and on the OpenSARShip-VI dataset, it increases by 6.80% from 47.71% to 54.51%. Significant improvements are achieved compared to the baseline network, indicating that guiding by target prototypes can suppress background speckle noise interference, making the representation of key area features more adequate. At the same time, compared to the baseline, the baseline-text method shows a substantial improvement, with an accuracy increase of 9.98% on the OpenSARShip-III dataset and 20.16% on the OpenSARShip-VI dataset. This fully demonstrates that adding text information promotes the description of discriminative features, significantly improving the recognition performance of the network. Notably, the performance improvement brought by the baseline-text, which only adds the text feature processing branch, is higher on both datasets than the baseline-prototype, which only adds the PGAM in the image feature processing. This proves the advantages of the image–text multimodal information processing paradigm over the traditional image processing paradigm in SAR ship recognition tasks.

Finally, as shown in the last row of Table V, PGMNet, which incorporates both target prototype guidance and image–text multimodal processing architecture, achieves the highest accuracy on both datasets, 88.07% and 72.34%, which are 14.59% and 24.63% higher than the baseline, respectively. It also surpasses the corresponding baseline-prototype and baseline-text methods, proving that the proposed TIUS can fully integrate image–text features, enhance intraclass feature consistency, increase interclass feature diversity, and achieve high precision in the task of SAR ship recognition.

2) Qualitative Analysis: As shown in Fig. 9, to more intuitively demonstrate the effectiveness of the designed modules, t-distributed stochastic neighbor embedding (t-SNE) [87] is selected for qualitative analysis on the seven-category FUSAR-Ship dataset. Fig. 9(a)–(d) represents the visualization results of the baseline, baseline-prototype, baseline-text, and PGMNet networks, respectively.

From Fig. 9(a), it can be seen that as the baseline network, ResNet50 exhibits significant confusion in feature embeddings. Except for the types of other cargoes and tankers, the interclass distances for other categories are small, and the intraclass features are dispersed. After introducing the target prototype guidance, as shown in Fig. 9(b), the feature embeddings of categories such as general cargo and tanker show significant improvement, with enhanced intraclass feature consistency and further increased interclass distances, indicating a certain improvement in recognition performance.

When only text features are added to the baseline, as shown in Fig. 9(c), the constraints of target size information effectively increase interclass differences. However, as shown in Table III, some confusion remains between size-similar categories in the FUSAR-Ship dataset, such as other and other cargo, and fishing and other. Nevertheless, categories with significant size differences, such as bulk carriers, container ships, and general cargo, clearly separate feature embeddings from other categories.

The results in Fig. 9(d) reflect the superior performance of PGMNet on the FUSAR-Ship dataset. Compared to Fig. 9(a)–(c), the feature embeddings of each category are more aggregated, and the interclass distances are more distinct. This effectively demonstrates PGMNet's ability to address the issues of large intraclass and small interclass differences in SAR ship recognition, better capture the intrinsic characteristics of targets, and achieve high-precision recognition.

E. Influence of Prototype and Key Area Generation Strategy

To further discuss the impact of the generation strategy of the target prototype and the key area on accurately extracting ship size information, we conduct the following experiments, as shown in Fig. 10. The process guided by the blue arrow involves the direct extraction of SSPs from SAR ship images and fitting size information, and the process guided by the yellow arrow involves extracting ship information from key areas corresponding to SAR ship images. Fig. 10(a)–(c), respectively, represents the effects of size information extraction using different methods for images with different resolutions in various scenarios.

In Fig. 10(a), two ship targets are present in a single SAR image. The direct extraction method captures the SSPs of both ships, and the resulting fit curve does not accurately reflect the true size of the targets. In this article, we use semantic segmentation to isolate the larger ship and extract its intrinsic information. This approach allows for a more accurate acquisition of text features representing the ship's size.

Fig. 10(b) shows a ship affected by sea clutter in the background. Here, directly extracting SSPs is severely interfered

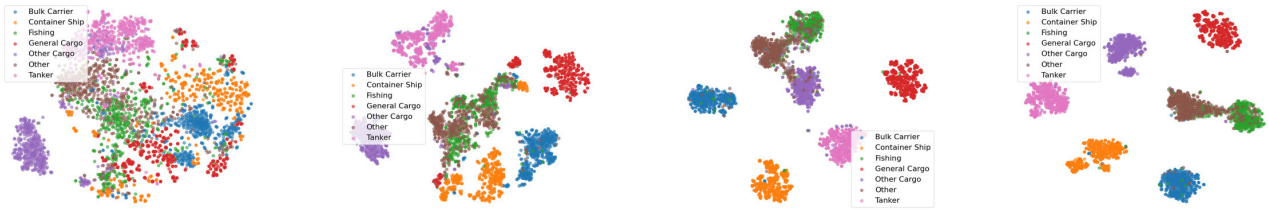


Fig. 9. Ablation experiments of PGMNet: t-SNE feature visualization analysis on the seven-category FUSAR-Ship dataset. (a) Baseline. (b) Baseline-prototype. (c) Baseline-text. (d) PGMNet.

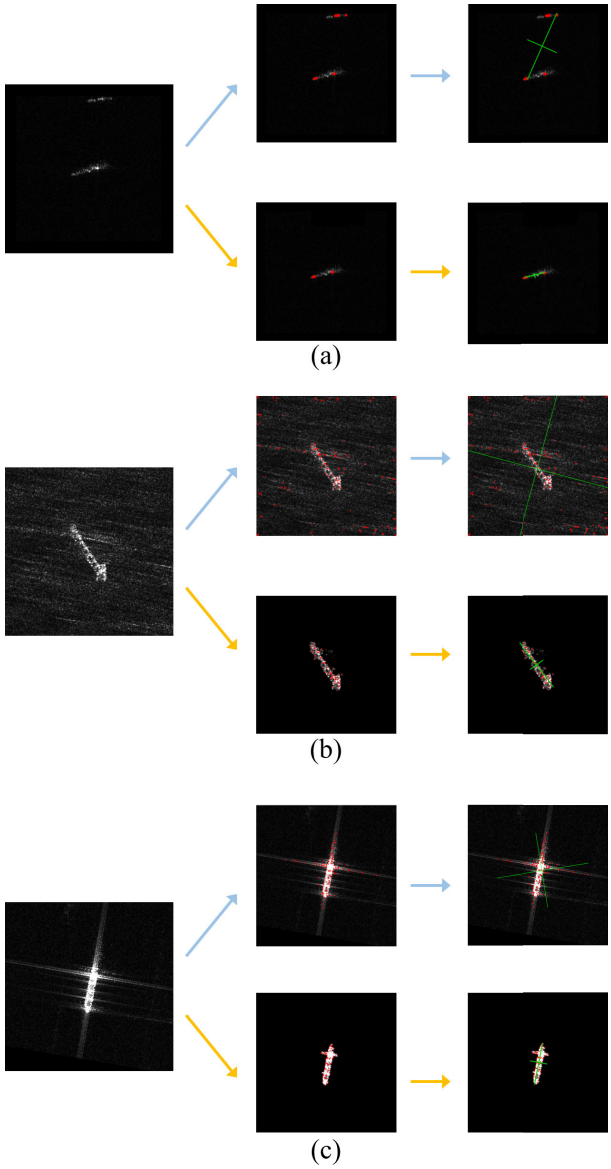


Fig. 10. Different processes for extracting ship size information. The blue arrow represents the process of directly extracting size information from original images, and the yellow arrow indicates the process of generating key areas first, followed by extracting ship size information from these areas. Red points denote the extracted SPPs, while green lines indicate the fitted ship size. (a) Two ship targets appear in the same image. (b) Shows interference from speckle noise. (c) Depicts the presence of cross-shaped bright spots on the ship.

with by the background, leading to inaccurate SSP extraction and ultimately failing to extract the correct size information. The proposed PGMNet filters out background clutter interference, thus effectively describing the SSPs in the key area and obtaining precise size information.

Due to the special imaging mechanism, the dihedral structures between the metal superstructures and the hulls of ships easily form cross-shaped bright spots, as shown in Fig. 10(c). Directly extracting SSPs from the original image would interfere with the bright spots, enlarging the real size of the ship. However, in this article, the prototype and key area generation strategy are employed to minimize the cross-shaped bright spots effects, thereby improving the accuracy of target size representation.

Fig. 10 effectively demonstrates that the designed target prototype and key area generation strategy can reduce the interference from complex backgrounds, special imaging conditions, and other factors, greatly improving the accuracy of target size information extraction. This will effectively enhance the consistency of the unified representation of image–text multimodal features.

F. Influence of Different Numbers of Training Samples

In practical applications, SAR ship datasets frequently suffer from a shortage of samples. To validate the performance of the proposed PGMNet under various settings of training samples in quantity, ample experiments are conducted on the OpenSARShip-VI dataset. Samples are randomly selected from 200 training samples per category, setting the number as 20, 40, 60, 80, 100, 120, 140, 160, 180, and 200, respectively. The performance in the test stage via trained models is detailed in Table VI, which includes the results of ResNet50 as the baseline and HDSSNet [71], the second-best performing method on the OpenSARShip-VI dataset, for comparison. To intuitively demonstrate the performance of the models, the experimental results are plotted in a bar chart, as shown in Fig. 11. In these experiments, the models labeled 20, 40, 60, 80, 100, 120, 140, 160, 180, and 200 correspond to PGMNet trained with 20, 40, 60, 80, 100, 120, 140, 160, 180, and 200 training samples per category, respectively.

Denoting the number of samples in each category during the training stage as N, Fig. 11 shows that when N is 20, recognition accuracy is relatively low. However, when it reaches 40, the performance of PGMNet improves significantly, achieving a recognition accuracy of 49.76%, surpassing the baseline ResNet50's 47.71%. With N reaching 60 during training, PGMNet obtains an accuracy of 63.09%, exceeding the performance of HDSSNet [71], which is trained with

TABLE VI
PERFORMANCE OF DIFFERENT MODELS ON THE OPENSARSHIP-VI DATASET. NOTE THAT THE BEST RESULTS ARE MARKED IN BOLD

Model \ Acc(%)	Bulk Carrier	Cargo	Container Ship	Fishing	General Cargo	Tanker	Average
20	**79.83**	15.85	23.47	68.00	28.57	39.08	31.76
ResNet50	58.37	40.81	58.99	92.00	20.24	20.77	47.71
40	43.35	46.06	49.68	88.00	48.81	69.01	49.76
HDSSNet	60.09	46.23	**84.04**	74.00	48.81	44.01	60.76
60	55.79	65.94	60.36	88.00	44.05	73.94	63.09
80	39.27	62.87	77.06	86.00	48.81	75.7	64.87
100	34.55	73.99	75.58	90.00	25.00	80.99	67.87
120	62.66	**83.54**	57.61	90.00	29.76	73.94	69.65
140	59.66	73.73	66.49	92.00	54.76	79.93	69.58
160	60.52	72.77	68.50	88.00	52.38	**82.75**	70.12
180	61.16	72.5	71.46	86.00	**64.29**	80.28	71.13
200	60.30	79.68	67.97	**96.00**	52.38	78.87	**72.34**

200 samples per category, reaching the SOTA performance on the OpenSARShip-VI dataset. As N increases sequentially, the accuracy of recognition steadily improves. The multimodal information fusion paradigm represented by PGMNet evidently achieves a higher performance ceiling compared with other methods.

Of course, from Table VI, although PGMNet outperforms the baseline when trained with 40 samples, there are still variations in recognition performance across different categories. For instance, lower accuracy is observed for general cargo and tankers. This issue likely stems from the large size variations of ships across different viewing angles and insufficient training samples, which impede the model's ability to learn robust discriminative features. These findings underscore that how to enhance recognition accuracy across all categories under limited training sample conditions remains an open question worthy of further investigation.

Generally speaking, the proposed guided image–text multimodal recognition network in this article can better extract discriminative features and achieve high performance with relatively few training samples. Compared to other deep learning models, it is more robust in recognizing SAR ship images and is demanding less number of samples in the training stage.

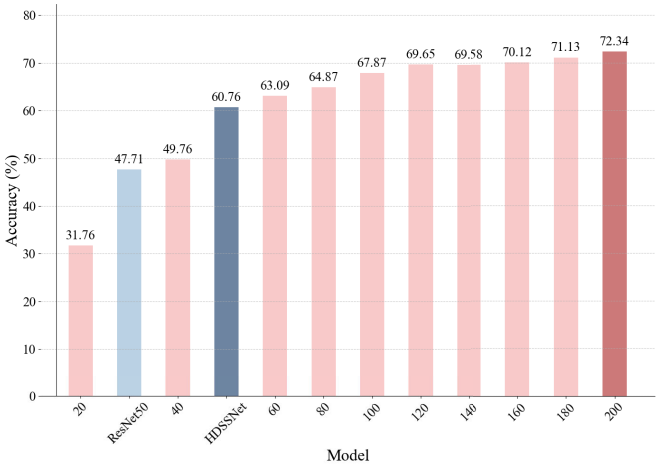


Fig. 11. Accuracy of different models on the OpenSARShip-VI dataset. In the figure, the horizontal axis values—20, 40, 60, 80, 100, 120, 140, 160, 180, and 200—represent the models obtained by training the PGMNet model with 20, 40, 60, 80, 100, 120, 140, 160, 180, and 200 samples per class, respectively. ResNet50 and HDSSNet serve as comparison networks, with 200 samples per class used for training.

G. Influence of Different Text Encoders

To further investigate the influence of text encoders on algorithm performance, we conducted comparative experiments using three text encoders with comparable parameter scales: GPT-2 [88], BERT [89], and CLIP [69], evaluated on the OpenSARShip and FUSARShip datasets. The baseline-prototype model (described in Section IV-D) represents the framework without any text encoder. The models "With-GPT," "With-BERT," and "With-CLIP" denote implementations using GPT-2, BERT, and CLIP (our proposed network PGMNet) as the text encoder, respectively. The detailed results are summarized in Table VII.

From Table VII, it is evident that integrating CLIP as the text encoder yields the best ship-recognition accuracy across all three datasets: 88.07% on OpenSARShip-III, 72.34% on OpenSARShip-VI, and 93.19% on FuSARShip, respectively. This represents improvements of 4.22%, 4.94%, and 1.15% over the next best model, which uses BERT as a text encoder. In contrast, GPT-2 exhibits the weakest performance among the three encoders. This performance discrepancy stems from differences in pretraining strategies and architectural designs. CLIP, pretrained through contrastive learning on massive image–text pairs, enables its text encoder to capture rich cross-modal semantic relationships, thereby enhancing generalization capability and facilitating effective image–text feature alignment and fusion. While BERT benefits from bidirectional contextual modeling, its text-only pretraining limits cross-modal alignment with visual features. GPT-2, as a unidirectional autoregressive model, prioritizes text generation during pretraining, resulting in less efficient global semantic representations compared to the other two encoders.

TABLE VII
SHIP RECOGNITION ACCURACY OF NETWORKS USING DIFFERENT TEXT ENCODERS ON THE OPENSARSHIP AND FUSAR-SHIP DATASETS. NOTE THAT THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Model	Accuracy(%)↑		
	OpenSARShip-III	OpenSARShip-VI	FUSAR-Ship
Baseline-Prototype	80.73	54.51	87.11
With-GPT	81.44	63.83	90.67
With-BERT	83.85	67.40	92.04
With-CLIP(PGMNet)	**88.07**	**72.34**	**93.19**

TABLE VIII

ABLATION EXPERIMENTS FOR PGMNET. NOTE THAT THE BEST RESULTS ARE MARKED IN BOLD

Model Fusion Strategy	Acc(%)↑	F1(%)↑
Baseline-Prototype	54.51	46.41
Addition	65.01	54.22
Concatenation	66.08	54.12
Multiplication	69.11	62.09
TIUS(PGMNet)	**72.34**	**66.67**

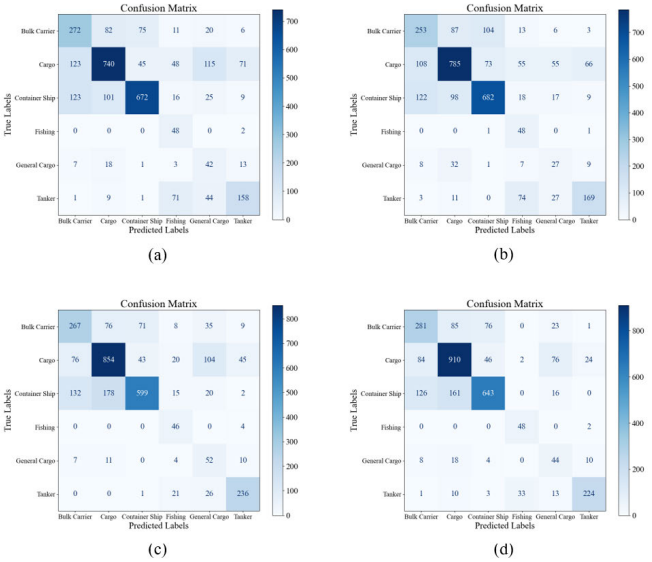


Fig. 12. Confusion matrix comparison of different multimodal feature fusion methods on the OpenSARShip-VI dataset. (a) Addition. (b) Concatenation. (c) Multiplication. (d) PGMNet.

Moreover, all text-feature-processing-branch-using models outperform the baseline-prototype (without a text branch), further validating the effectiveness of the proposed image–text fusion paradigm for SAR ship recognition.

H. Influence of Different Fusion Strategies

To further verify the efficiency of the proposed TIUS in PGMNet, as depicted in Fig. 6(a)–(c), we conduct a comparison between our network and other fusion methods such as addition, concatenation, and multiplication. These methods are experimentally validated on the OpenSARShip-VI dataset, with their results presented in Table VIII. To ensure a fair comparison, the image feature processing parts of the other three fusion models are kept the same as in PGMNet, differing only in the image–text feature fusion approach. By the way, in Table VIII, we also include baseline-prototype, which only processes image features and does not perform any fusion, to serve as a baseline for evaluating the fusion performance. In addition, we utilize confusion matrices to further analyze and compare the impact of different fusion methods on SAR ship recognition, as shown in Fig. 12.

As shown in Table VIII, after introducing text features, all four multimodal feature fusion methods achieve performance improvements. It is clear that PGMNet achieves the best recognition performance with an accuracy of 72.34% and an $F1$ of 66.67%, which is significantly higher than the other three fusion methods. This demonstrates that simple fusion methods-such as addition, concatenation, and multiplication are inadequate for effectively integrating image and text features exhibiting significant domain differences. The TIUS proposed in PGMNet leverages MLP structures to map image and text features into a high-dimensional space, allowing for a more comprehensive fusion of multimodal features and enhancing the extraction and representation of discriminative features.

TABLE IX

COMPARISON OF MODEL ACCURACY, PARAMETER COUNT, AND FLOPS. NOTE THAT THE BEST RESULTS ARE MARKED IN BOLD

Model	Acc(%)↑	Params(M)↓	FLOPs(G)↓
AlexNet	40.38	58.31	**1.13**
VGG-16	48.76	134.29	15.47
ResNet-50	47.71	23.52	4.13
DenseNet-161	52.49	26.49	7.84
Inception-v3	44.21	21.8	6.38
Inceptionnn-v4	44.78	41.15	13.72
Xceptionn	51.11	20.82	9.41
Wang et al.	50.53	15.52	15.35
Zeng et al.	55.26	237.03	15.57
SE-LPN-DPFF	56.66	**4.22**	36.53
HOG-ShipCLSNet	53.77	100.23	2.96
HDSSNet	60.76	48.785	5.62
PGMNet(Ours)	**72.34**	109.92	5.24

The confusion matrices in Fig. 12 further illustrate the advantages of TIUS over conventional fusion algorithms. This strategy enables PGMNet to significantly enhance the intraclass feature consistency of cargo and tanker categories, greatly reducing their confusion with other categories. This highlights the importance of efficient multimodal feature fusion methods in addressing the issue of large intraclass diversity and interclass similarity in SAR ship recognition.

I. Analysis on Floating Point Operations (FLOPs) and Params

In addition to showing the accuracy of these SOTA algorithms on the OpenSARShip-VI dataset, Table IX also compares their parameters and FLOPs. Except for VGG-16, classical CNN-based models generally have fewer parameters and lower FLOPs. As the parameters and FLOPs increase, the performance of networks designed specifically for SAR ship recognition also improves. As an image–text multimodal information fusion network, PGMNet has more parameters than pure image processing frameworks. However, compared to HOG-ShipCLSNet [11], which incorporates additional information, PGMNet maintains a comparable parameter count while delivering superior SAR ship recognition performance. Notably, PGMNet significantly outperforms Zeng et al. [84] in recognition accuracy while requiring fewer parameters.

V. CONCLUSION

To address the issues of background sea clutter interference in SAR ship images, the PGAM is proposed to focus on and utilize key area information through the guidance of target prototypes. Furthermore, to better leverage multimodal information, a text feature processing branch is designed to extract and integrate multimodal features efficiently. The proposed PGMNet introduces an image–text multimodal fusion

paradigm into SAR ship recognition, breaking through the performance bottleneck associated with traditional image-only processing frameworks. This approach effectively mitigates the challenges of large intraclass diversity and interclass similarity in SAR ship recognition, significantly enhancing recognition accuracy. Extensive experiments demonstrate that the proposed PGMNet achieves SOTA performance, with recognition accuracy improvements of 4.84%, 11.58%, and 2.20% over the current best-performing algorithms on the OpenSARShip-III, OpenSARShip-VI, and FUSAR-Ship datasets, respectively.

Experiments indicate that introducing the prototype and key area generation strategy successfully incorporates semantic segmentation concepts into SAR image target recognition, ensuring efficient extraction of key area information and accurate acquisition of ship size information. Furthermore, the designed PGAM based on the special imaging mechanism of SAR effectively enhances the intraclass feature consistency representation in SAR images. Experiments with different multimodal feature fusion methods also confirm that the proposed TIUS excels in fusing image–text features with significant domain differences, greatly enhancing multimodal feature representation. Moreover, PGMNet has been proven to achieve excellent performance with a few samples, demonstrating its potential for practical applications.

There are still some limitations in our work. First, due to the special imaging mechanism of SAR, SAR ships exhibit imaging variability, causing fluctuations in the extracted ship size information, particularly when training samples are limited, which might decrease model performance. In addition, as PGMNet is a multimodal recognition network, it involves extracting and representing both image and text features, resulting in a larger number of parameters than traditional image-only processing networks.

Our future work is summarized as follows.

1) *Enhanced Robustness:* We aim to leverage electromagnetic simulation techniques to comprehensively capture the imaging characteristics of SAR ships from different angles, obtaining complete prior information to train a more robust multimodal SAR ship recognition network.
2) *Optimization and Efficiency:* We aim to reduce the network's parameters with minimal loss in performance, achieving an efficient balance between parameter quantity and algorithm accuracy.
3) *Broader Application:* We intend to extend the image–text multimodal information fusion paradigm to recognize other SAR image targets, such as vehicles and aircraft. In addition, we will explore the implementation and effectiveness of combining text features in multimodal processing paradigms for SAR image detection tasks.

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